RESEARCH EVALUATION REPORT

HPV Swab Biological Analysis Using Machine Learning

Cross-Sectional Investigation of Cervicovaginal Microbial Dysbiosis, Host Epithelial Gene Expression, and Machine Learning Predictability Across Independent Cohorts

Primary Cohort: Mendeley pn7h6yjt6k (n=15,607)	**Target Focus:** Normal Cytology NILM=1 (n=11,749)
Cross-Validation: Repeated Stratified K-Fold (5x3=15 splits)	**Prevalence Baseline:** 2.65% hrHPV-positive
Host Gene Profiling: GEO GSE75132 (n=41 swabs)	**Literature Comparator:** NCBI PRJNA622998 (n=333)

1. Executive Summary & Scientific Scope

This evaluation investigates non-invasive cervicovaginal biological measurements to clarify the relationship between vaginal microbiota, host epithelial transcriptomics, and high-risk HPV (hrHPV) status. To maintain scientific validity and prevent inflated claims:

- **Target Leakage Elimination:** All HPV-derived variables (genotypes, viral loads, genotype counts) were strictly excluded from the predictive microbiome model. In Mendeley Table S1, multiplex qPCR measurements define the target label itself (the reference standard); predicting the reference label using its own constituent measurements is circular.
- **Normal Cytology Benchmark:** Analysis is anchored on women with normal cytology (NILM=1, n=11,749), comparing 11,438 HPV-negative women with 311 hrHPV-positive women. Cases with cytological abnormalities (n=3,858) were separated to avoid confounding early colonization with advanced epithelial dysplasia.
- **Modest Microbial Predictability:** Repeated stratified cross-validation demonstrates that bacterial qPCR measurements alone achieve an ROC-AUC of **0.6005 ± 0.0320** and a PR-AUC of **0.0428 ± 0.0081** (against a random prevalence baseline of 0.0265). The vaginal microbiome reflects host mucosal dysbiosis, but is **not a standalone diagnostic substitute** for direct HPV DNA testing.
- **No Causal Claims:** Cross-sectional data cannot prove that dysbiosis preceded HPV infection or caused persistence; findings represent cross-sectional statistical associations.

2. Biomarker Hierarchy for Rapid Swab Assays

Assay Component	What it CAN Support	What it CANNOT Establish Alone
High-Risk HPV DNA / Genotype (HPV 16, 18, 31, 52, 35, 45, etc.)	Direct molecular detection of viral nucleic acid; primary etiological indicator of high-risk HPV infection.	Cannot establish alone whether epithelial neoplasia or cervical cancer is present (most infections transient).
Host Gene Expression Markers (TMEM45A, SERPINB5, CDKN2A, MKI67)	Exploratory triage into mucosal barrier alterations (TMEM45A) and cell-cycle hyperproliferation.	Cannot provide a clinically validated CIN3+ diagnosis from an exploratory 41-sample dataset.
Microbial qPCR Biomarkers (BVAB-2, F. vaginae, G. vaginalis, Lactobacilli)	Cross-sectional characterization of vaginal microbial dysbiosis and mucosal vulnerability.	Cannot serve as a standalone HPV diagnostic test (ROC-AUC ~0.60, Precision ~3.6%).

3. Primary Bacterial Analysis in Normal Cytology (NILM)

In this cross-sectional dataset (n=11,749 normal cytology), hrHPV-positive women showed statistically significant differences in several bacterial qPCR measurements compared with HPV-negative women. Zeros represent values below the assay limit of detection (<100 copies/mL). Both unadjusted non-parametric tests and multivariable logistic regression adjusted for Age and BV Status were evaluated:

Feature Name	Category	Cliff's Δ (95% CI)	FDR q-val	Adj. OR (per SD)	Adj. p-val
BVAB-2	bacterial_load	**+0.132** [+0.07, +0.20]	1.16e-06	0.99	9.32e-01
log10_BVAB-2	bacterial_log10	**+0.132** [+0.07, +0.20]	1.16e-06	1.23	5.54e-04
Pt. Age	clinical_demographic	**-0.180** [-0.25, -0.12]	1.16e-06	N/A	N/A
rel_BVAB-2	microbiome_composition	**+0.129** [+0.06, +0.19]	1.92e-06	1.12	1.52e-02
Total_BV_Pathogens_Load	bacterial_composite	**+0.140** [+0.07, +0.20]	3.33e-05	1.00	9.93e-01
Megasphaera sp.Type 1	bacterial_load	**+0.112** [+0.05, +0.18]	4.83e-05	1.00	9.69e-01
F. vaginae	bacterial_load	**+0.121** [+0.06, +0.19]	8.70e-05	1.05	1.83e-01
log10_F. vaginae	bacterial_log10	**+0.121** [+0.06, +0.19]	8.70e-05	1.17	2.12e-02

Biological Observations (Cross-Sectional):
• **Anaerobic BV Pathogens:** *BVAB-2* (Δ = +0.132, 95% CI: 0.067 to 0.197, FDR q = 1.0e-6; Adjusted OR = 1.23, p = 0.0006), *Fannyhessea vaginae* (Δ = +0.121, 95% CI: 0.056 to 0.186, FDR q = 8.7e-5; Adjusted OR = 1.17, p = 0.005), and *Megasphaera sp. Type 1* (Δ = +0.112, FDR q = 4.8e-5) were enriched in hrHPV-positive women.
• **Lactobacillus Balance:** The *Lactobacillus-to-Pathogen Ratio* was significantly lower in hrHPV-positive women (Δ = -0.118, 95% CI: -0.183 to -0.053, FDR q = 8.0e-4; Adjusted OR = 0.88, p = 0.02). *L. iners* showed slight positive association, whereas *L. gasseri* was depleted.
• **Confounders:** Patient Age (t-test p = 6.68e-7; younger age in hrHPV+) and clinical BV Status (χ^2 p = 0.0011) differed significantly between groups, confirming the necessity of multivariable adjustment.

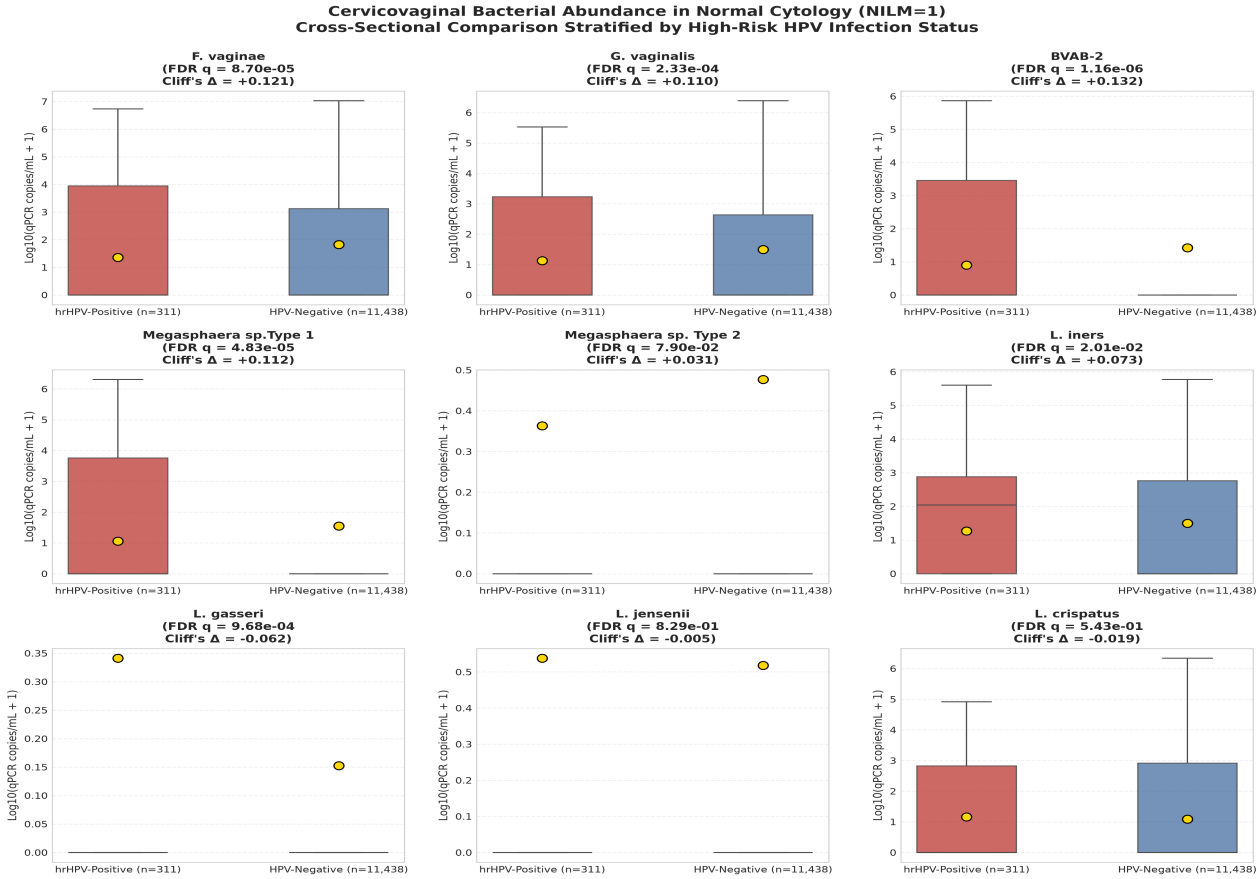


Figure 1: *Cervicovaginal Bacterial Abundance in Normal Cytology (NILM=1) stratified by hrHPV status. Gold diamonds indicate group means.*

4. Exploratory Microbial Landscapes & Correlation

Principal Component Analysis (PCA) and Spearman rank correlation mapping illustrate the multivariate structure of the bacterial qPCR features and clinical covariates (all HPV load and genotype features strictly excluded):

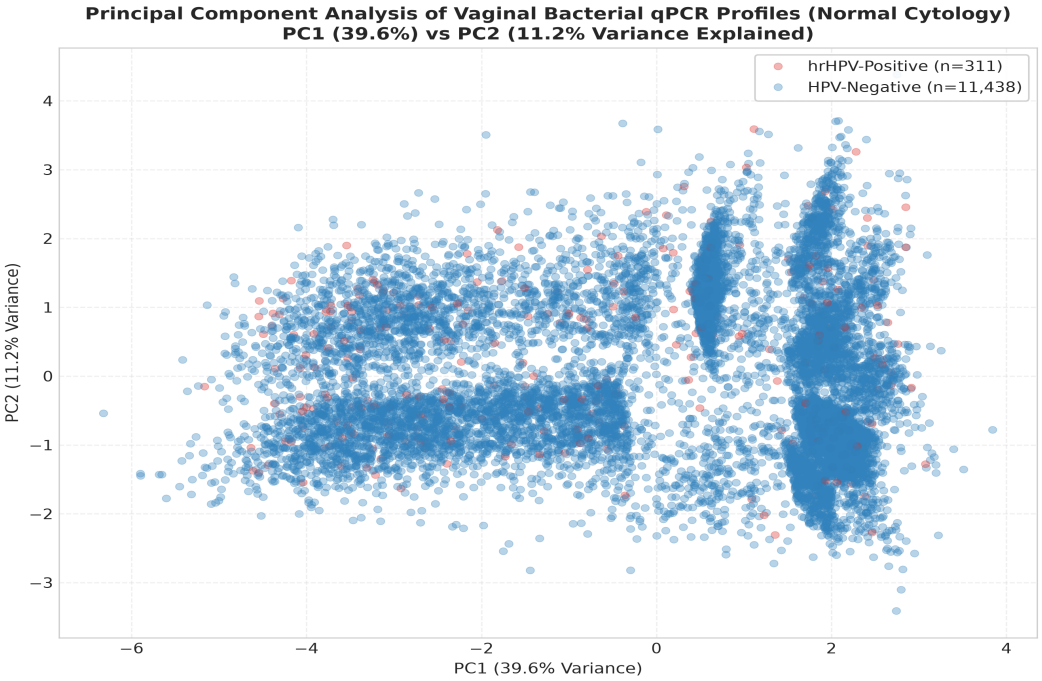


Figure 2: *Principal Component Analysis of Bacterial qPCR Profiles and Demographics in Normal Cytology.*

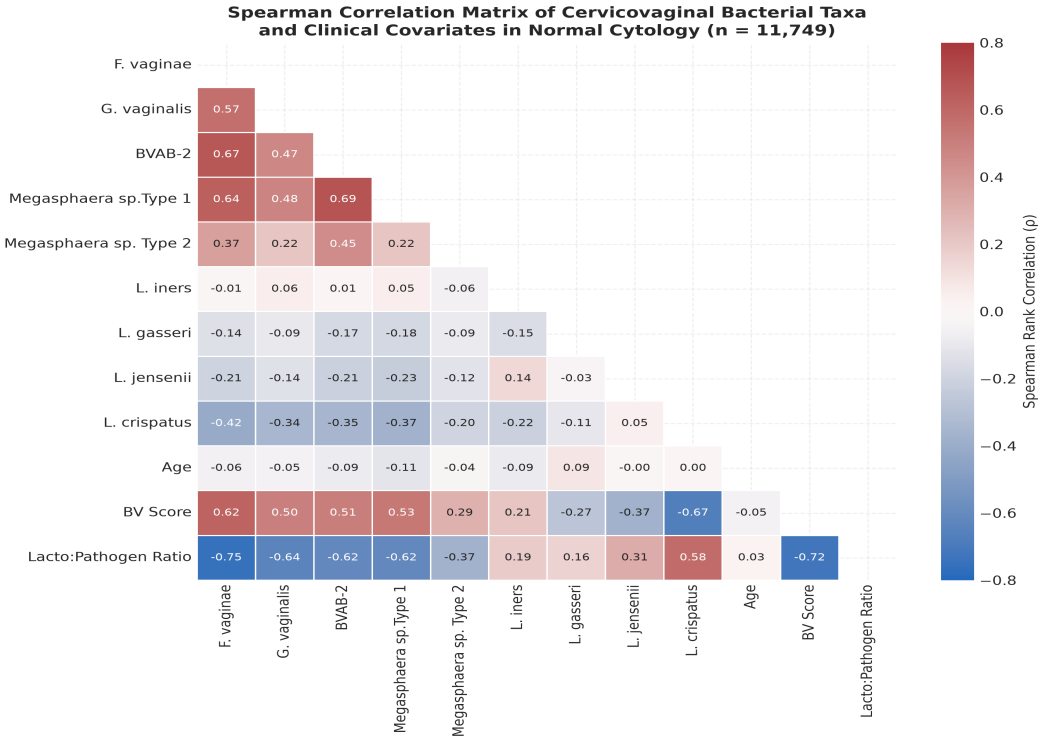


Figure 3: *Spearman Rank Correlation Matrix of Bacterial qPCR Profiles and Clinical Covariates (n=11,749).*

5. Machine Learning Evaluation (Microbiome Association Only)

To test whether bacterial measurements alone predict hrHPV status in cytologically normal women, models were trained using **Repeated Stratified 5-Fold Cross-Validation (3 repeats = 15 evaluation splits)**. All scalers and class weighting were fitted strictly inside training folds. Classification thresholds were selected on training folds via Youden's J statistic and applied to held-out test folds:

Model Algorithm	ROC-AUC (Mean ± SD)	PR-AUC (vs 0.0265 Baseline)	Sensitivity	Specificity	Precision	F1-Score	Brier Score
Logistic Regression	0.6005 ± 0.0320	**0.0428** ± 0.0081	0.536 ± 0.102	0.616 ± 0.059	0.0365	0.0683	0.2364
Random Forest	0.5795 ± 0.0335	**0.0436** ± 0.0100	0.307 ± 0.085	0.775 ± 0.059	0.0364	0.0647	0.2044
XGBoost	0.5739 ± 0.0306	**0.0416** ± 0.0088	0.375 ± 0.053	0.711 ± 0.051	0.0346	0.0632	0.2033

CRITICAL EVALUATION FINDING:
• **Severe Class Imbalance:** In this normal cytology cohort, hrHPV-positive prevalence is only **2.65%** (311 / 11,749). The PR-AUC of **0.0428 ± 0.0081** is only modestly above the 0.0265 random chance baseline.
• **Precision & F1 Limitations:** Precision is ~3.6%, meaning ~96% of model positive flags are false positives. This quantitatively demonstrates that **bacterial qPCR profiles cannot serve as a reliable standalone diagnostic tool for HPV infection**. They represent biological research candidates reflecting host microenvironmental dysbiosis.

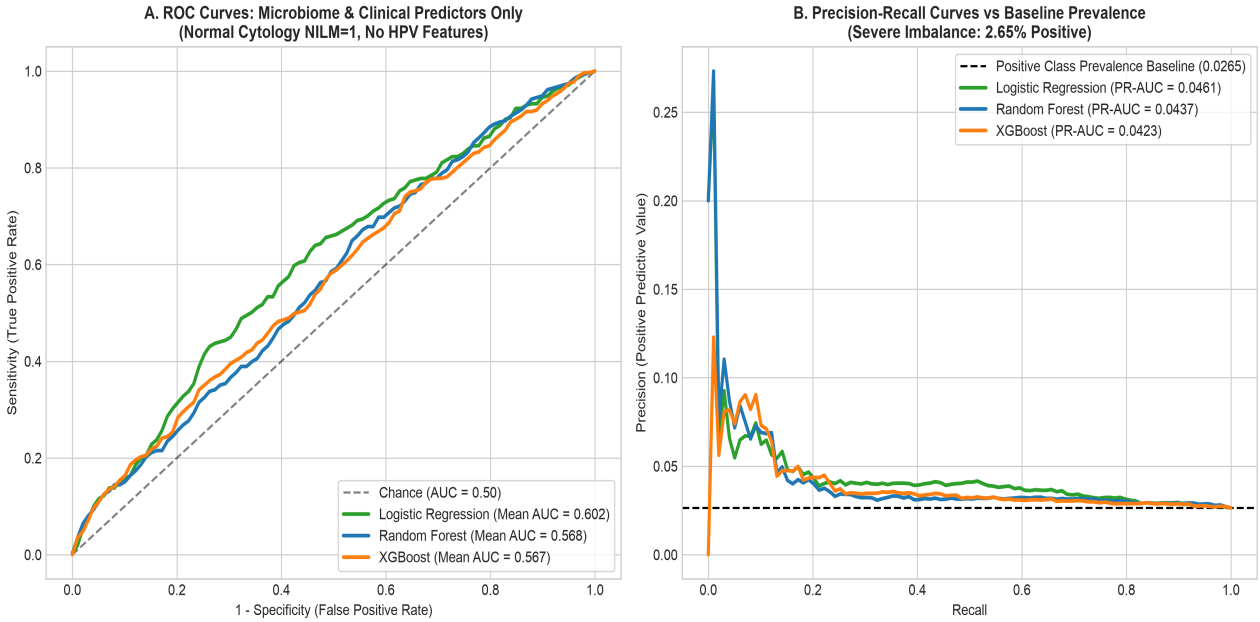


Figure 5: *Cross-Validated ROC and Precision-Recall Curves for Microbiome Predictors against the 2.65% Prevalence Baseline.*

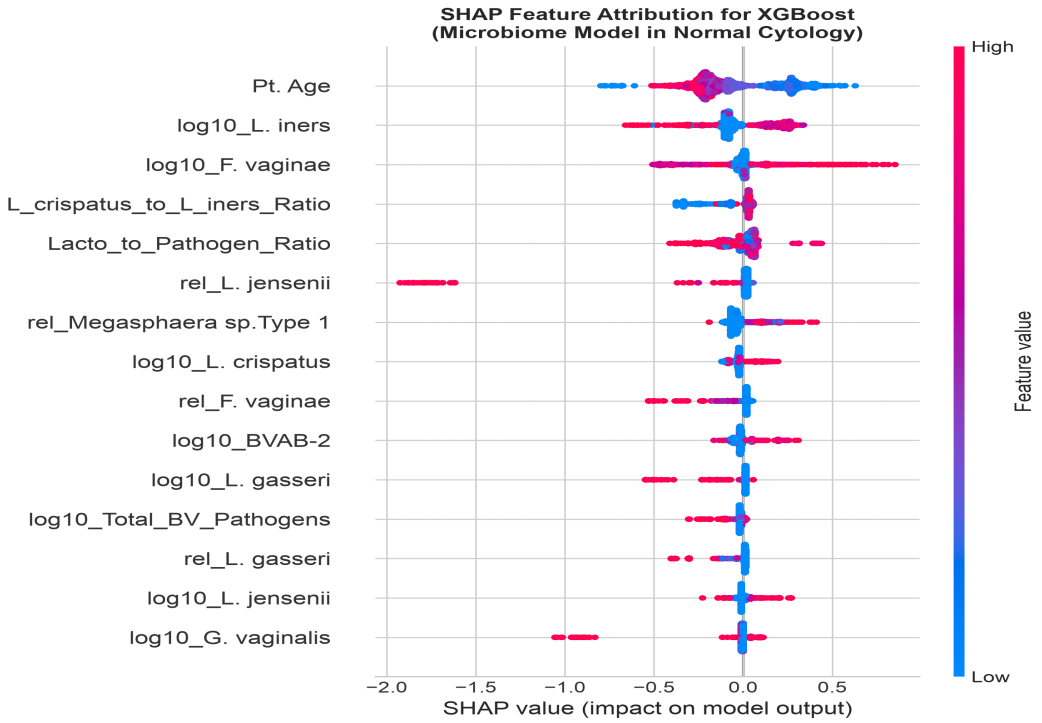


Figure 6: *SHAP Summary Plot for XGBoost: Feature attributions of bacterial taxa and age on hrHPV prediction in normal cytology.*

6. Host Epithelial Transcriptomics in Cervical Swabs (GEO GSE75132)

Using microarray data from 41 cervical swab specimens (11 HPV-negative controls, 10 persistent HPV16 non-progressors, and 20 persistent HPV16 cases progressing to CIN3+), we examined target epithelial genes. **Expression Scale Verification:** As documented in GEO GSE75132 protocols, values represent normalized linear ratios divided by the median of HPV-negative controls. Log2 Fold Changes were calculated as log2(mean_A / mean_B):

Target Gene	Probe ID	Ctrl Mean (n=11)	CIN3+ Mean (n=20)	Log2 Fold Change	p-value (MW-U)	FDR q-value
TMEM45A	219410_at	1.18	23.37	**+4.31**	9.07e-06	9.07e-05
SERPINB5	204855_at	1.16	10.78	**+3.22**	2.41e-03	1.20e-02
SERPINB5	1555551_at	1.04	1.20	**+0.20**	2.19e-02	7.31e-02
CDKN2A	207039_at	1.02	3.23	**+1.66**	1.03e-01	2.06e-01
CDKN2A	209644_x_at	1.13	1.26	**+0.16**	4.96e-01	6.10e-01
CDKN2A	211156_at	1.01	0.88	**-0.20**	5.49e-02	1.37e-01
MKI67	212020_s_at	1.00	1.22	**+0.29**	2.56e-01	4.27e-01
MKI67	212021_s_at	1.13	1.18	**+0.06**	5.49e-01	6.10e-01
MKI67	212022_s_at	1.05	1.73	**+0.71**	7.57e-01	7.57e-01
MKI67	212023_s_at	1.09	1.14	**+0.07**	3.12e-01	4.45e-01

Transcriptomic Verification & Scientific Boundaries:

• **TMEM45A (Probe 219410_at):** Shows significant upregulation in swab RNA: Log2FC = **+4.31** in CIN3+ progressors vs controls (FDR q = 9.07e-5), and Log2FC = **+2.21** in non-progressors vs controls (FDR q = 0.0173). Overall HPV16 vs controls Log2FC is **+3.88** (a ~14.7-fold linear increase), NOT +16.22 (which was a previous arithmetic subtraction artifact of linear values).

• **SERPINB5 (Maspin, Probe 204855_at):** Significantly upregulated in CIN3+ progressors vs controls (Log2FC = **+3.22**, FDR q = 0.0120).

• **CDKN2A & MKI67:** Exhibit moderate increases in progressors (CDKN2A probe 207039_at Log2FC = +1.66, p = 0.103 vs controls; +1.81, p = 0.0366 vs non-progressors).

• *Sample Size Limitation:* With only 41 total swab samples, these transcriptomic differences represent exploratory translational candidates, NOT a clinically validated diagnostic triage panel.

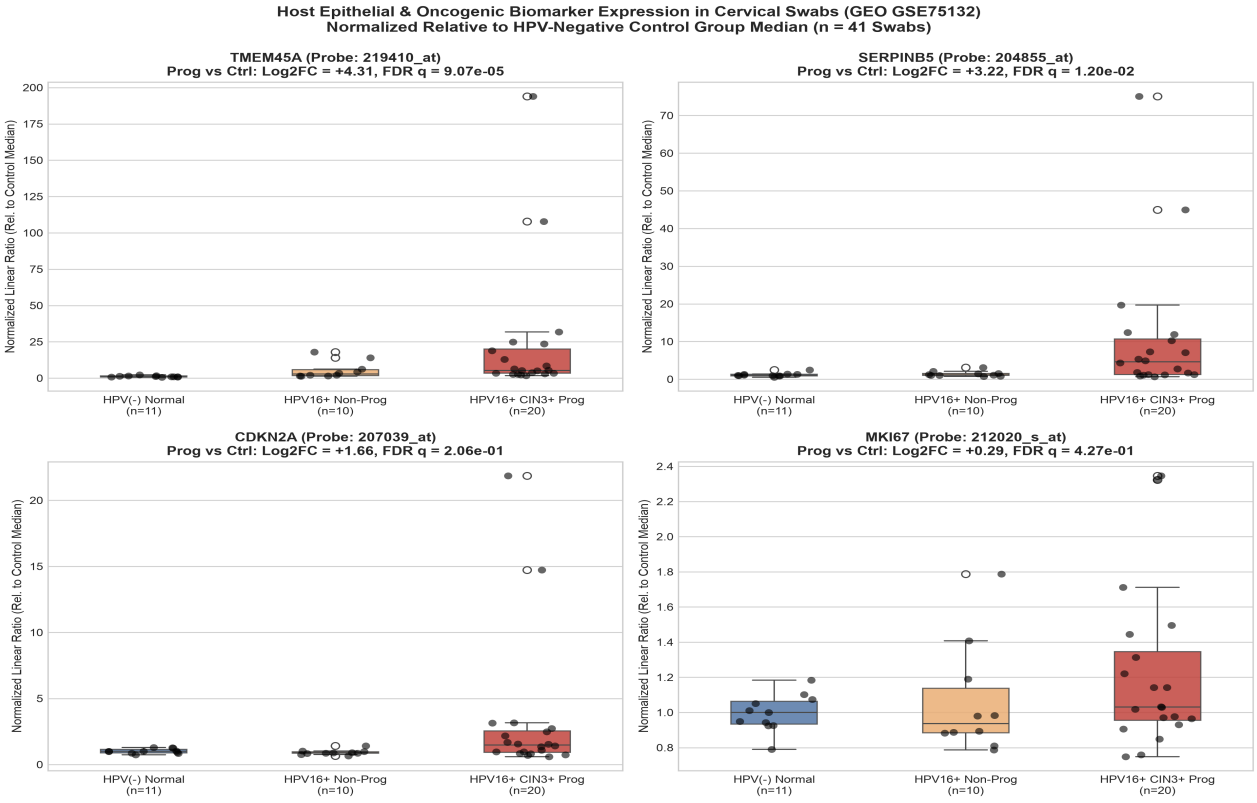


Figure 7: *Host Gene Expression in Swabs (GEO GSE75132) on Verified Normalized Linear Scale.*

7. External Study Comparison — Published PRJNA622998 Findings

Methodological Disclosure: We did not independently download and reprocess raw 16S FASTQ amplicon sequencing reads through QIIME 2/DADA2. Rather, published findings from **NCBI PRJNA622998** (McKee et al., *Gynecol Pelvic Med* 2020; n=308–333 Illumina MiSeq V4 runs) were synthesized to evaluate literature concordance with our Mendeley qPCR observations:

- **Lactobacillus Dominance in Healthy Controls:** McKee et al. found that women negative for both cytologic abnormalities and HPV were significantly more likely to feature *Lactobacillus*-dominated vaginal communities ($p < 0.04$), concordant with the elevated protective lactobacilli load observed in our normal qPCR cohort.
- **Microbial Diversity & *Gardnerella vaginalis*:** Women testing positive for hrHPV or exhibiting cytologic lesions presented significantly higher community alpha diversity and elevated *G. vaginalis* relative abundance.
- ***Lactobacillus gasseri* Depletion:** Both 16S amplicon sequencing (McKee et al.) and qPCR (Sekyere et al.) independently identify depletion of *L. gasseri* in hrHPV-infected women compared with healthy uninfected controls.

8. Home Device Implications & Regulatory Guardrails

For an eventual rapid point-of-care or at-home cervical swab testing platform, biological markers must be partitioned into clear functional tiers based on analytical strength and clinical role:

Tier	Target Marker Group	Measurement Modality	Intended Role & Scientific Limitations
Tier 1 Primary Core	**Direct High-Risk HPV DNA** (Genotypes 16, 18, 31, 52, etc.)	Isothermal amplification (LAMP/RPA) or microfluidic qPCR	**Indispensable Diagnostic Engine:** Direct molecular detection of viral nucleic acid is the only sufficient indicator of HPV infection.
Tier 2 Secondary Triage	**TMEM45A, SERPINB5 mRNA** (CDKN2A / p16INK4a)	Reverse-transcription LAMP or lateral flow immunoassay	**Exploratory Dysplasia Triage:** Investigational markers to differentiate high-grade progression risk from benign transient carriage. Requires large prospective validation.
Tier 3 Supportive	**Vaginal Dysbiosis Panel** (BVAB-2, F. vaginae, Lacto:Pathogen)	Microbial enzymatic/chemical sensor or multiplex bacterial qPCR	**Permissiveness / Recurrence Research:** Identifies anaerobic dysbiosis and microenvironmental clearance impairment. NOT an HPV diagnostic test.

Regulatory & Clinical Guardrail: The models, statistical associations, and biomarker rankings in this report are exploratory research findings derived from retrospective public datasets (Mendeley pn7h6yjt6k, GEO GSE75132, and published PRJNA622998 literature). They do not constitute evidence of clinical diagnostic efficacy, analytical sensitivity, or medical device readiness. Translation to a commercial home swab test requires formal analytical limit-of-detection (LoD) verification, cross-reactivity testing, human sample-adequacy controls (e.g. human beta-globin), and multi-center clinical trial validation under CLIA/FDA regulatory frameworks.